

Simulation Purpose & Scope Standard - (SPSS)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Operational Layer: Simulation Purpose & Scope Governance Layer

Category: Governance & Enforcement

Subcategory: Simulation Governance

Type: Parent Standard

Governed Space: Simulation Purpose & Scope

Version: 1.0

Status: Canonical · Open Standard

Effective Date: 16 August 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Simulation Purpose & Scope

Canonical Definition

Simulation Purpose & Scope Standard (SPSS) defines the governance conditions under which the reason for performing a simulation, the questions the simulation is intended to address, the portions of the governed Simulation Object to which it applies, the conditions and contexts included within the simulation inquiry, the material exclusions and limitations surrounding that inquiry, and the required simulation depth and precision are formally established before downstream simulation representation begins.

SPSS ensures that a technically successful simulation cannot silently acquire a broader purpose, scope, applicability, or interpretive meaning than was actually established before simulation design and execution.

Core Question

Why is the simulation being performed, what exactly must it examine, what remains outside its governed scope, and at what depth and precision must the simulation operate for its intended purpose?

Purpose

The purpose of SPSS is to transform a governed Simulation Object into a governed simulation inquiry.

SOS establishes:

What exactly is being simulated?

SPSS establishes:

Why is it being simulated and what must the simulation actually cover?

The distinction is fundamental.

The same Simulation Object may be simulated for radically different purposes.

A spacecraft may be simulated for: thermal behavior, navigation, structural stress, docking, autonomous control, communication loss, mission failure, or complete mission performance.

The Simulation Object may remain the same while the required simulation scope, depth, precision, conditions, and exclusions change substantially.

SPSS governs those differences.

Operational Role

SPSS is the second Parent Standard of SIMULOS® — Simulation Governance Architecture.

It receives the governed Simulation Object produced by SOS and establishes the simulation-specific purpose and scope required before the object can be represented inside a simulation.

Its role is to prevent: undefined simulation purpose, purpose substitution, scope ambiguity, hidden scope expansion, hidden scope contraction, partial simulation represented as whole-object simulation, excluded conditions being treated as simulated, inappropriate generalization of simulation outputs, insufficient simulation depth, unjustified precision, and downstream simulation design becoming detached from the question the simulation was originally intended to answer.

Governance Flow

Governed Simulation Object → Simulation Purpose → Simulation Questions → Simulation Scope → Scope Inclusions → Scope Exclusions → Applicability Context → Simulation Depth → Simulation Precision → Purpose & Scope Record → Governed Simulation Inquiry

The governed Simulation Purpose & Scope becomes the input to:

Reality Representation Governance

Simulation Purpose

Simulation Purpose defines why simulation is being performed and what decision, inquiry, exploration, comparison, investigation, design activity,

operational question, or knowledge requirement it is intended to support.

A Simulation Purpose may include: design exploration, performance assessment, behavior analysis, mission planning, safety investigation, failure exploration, scenario examination, stress assessment, training, operational preparation, comparative analysis, system optimization, autonomous-system testing, policy exploration, scientific inquiry, forecasting support, or another explicitly governed simulation objective.

The purpose must be sufficiently explicit to guide downstream simulation governance.

Purpose Must Be Object-Bound

The Simulation Purpose must remain connected to the governed Simulation Object.

A simulation purpose should therefore answer:

What are we trying to learn, examine, reproduce, compare, explore, or understand about this Simulation Object?

A purpose that cannot be connected to the governed object is not sufficiently established.

Primary and Secondary Purpose

A simulation may have more than one legitimate purpose.

SPSS may distinguish:

Primary Simulation Purpose

from:

Secondary Simulation Purposes

where necessary.

Secondary purposes must not silently become equivalent to the primary purpose if they require different representation, scenarios, depth, or precision.

Purpose Substitution

Purpose Substitution occurs when a simulation performed for one purpose is later treated as though it had been performed for another materially different purpose.

For example:

Design Exploration

must not automatically become:

Operational Safety Demonstration

and:

Training Simulation

must not automatically become:

Real-World Performance Evidence

without a governed basis.

SPSS prohibits silent purpose substitution.

Simulation Question

A Simulation Purpose should be translated into one or more sufficiently clear Simulation Questions.

Examples may include:

How does the system behave under condition X?

What happens if component Y fails?

How does the object respond across operating range Z?

Which scenario produces the greatest system stress?

How does autonomous behavior change when communication is lost?

The Simulation Question provides a bridge between broad purpose and concrete simulation scope.

Simulation Scope

Simulation Scope defines the governed extent of the simulation inquiry.

It determines which portions, characteristics, behaviors, conditions, interactions, operating modes, temporal periods, environments, populations, scenarios, and relationships are intended to be examined.

Simulation Scope establishes:

what the simulation covers

and:

what it does not cover. Object Boundary $\neq$ Simulation Scope

SIMULOS® establishes an important distinction:

Simulation Object Boundary $\neq$ Simulation Scope

SOS may define an entire aircraft as the Simulation Object.

SPSS may define the simulation scope as:

Landing Behavior Under Crosswind Conditions

The object remains the aircraft.

The simulation does not automatically cover every possible aircraft behavior.

Scope Inclusion

Material inclusions should be sufficiently explicit.

Depending upon the simulation, scope may include: specific components, functions, behaviors, operating modes, mission phases, environmental conditions, failure conditions, populations, interactions, geographic regions, temporal periods, or consequence pathways.

Scope Exclusion

Material exclusions must remain visible.

An exclusion identifies a condition, characteristic, behavior, environment, population, operating mode, component, or interaction intentionally outside the governed simulation inquiry.

SPSS establishes:

Excluded from Simulation Scope ≠ Demonstrated by Simulation

This distinction must remain preserved throughout downstream interpretation.

No Hidden Scope Expansion Rule

A simulation must not acquire broader scope after execution merely because the resulting outputs appear useful beyond the original inquiry.

For example:

Simulated urban daytime driving

must not silently become:

simulated autonomous driving under all operational conditions.

Any material expansion requires explicit reassessment of Simulation Purpose & Scope and affected downstream SIMULOS® spaces.

No Hidden Scope Contraction Rule

Likewise, difficult, adverse, or inconvenient conditions must not be silently removed from the Simulation Scope after they were established as materially relevant.

Scope contraction must remain explicit and its effect on downstream simulation claims must be preserved.

Scope Inflation

Scope Inflation occurs when a narrow simulation inquiry becomes represented as supporting a substantially broader conclusion.

Examples include:

Single Scenario → Complete Scenario Coverage

One Environment → All Environments

One Operating Mode → Complete Operational Envelope

One Population → General Population

One Mission Phase → Entire Mission

SPSS prohibits scope inflation without a governed basis.

Scope Fragmentation

A complex Simulation Scope may be divided into smaller governed simulation scopes.

This may be appropriate where different: environments, operating modes, components, scenarios, populations, or mission stages

require different simulation methods or representations.

Fragmentation must preserve the relationship between each partial scope and the complete intended inquiry.

Partial Simulation Scope

A partial scope is legitimate.

SIMULOS® does not require every simulation to cover the complete Simulation Object or complete operational reality.

It requires partial scope to remain explicit.

Therefore:

Partial Simulation ≠ Incomplete Governance

provided the boundary of the partial inquiry is clearly established.

Simulation Applicability Context

Simulation Applicability Context defines the conditions within which the simulation is intended to have meaning.

It may include: operational environment, geographic context, user population, mission phase, deployment condition, system state, autonomy level, interaction pattern, resource condition, consequence context, or another relevant domain.

Applicability Context is not equivalent to later validation or operational authorization.

It defines the context the simulation itself is intended to examine.

Simulation Purpose vs Operational Reliance

SPSS does not determine whether simulation outputs may be relied upon operationally.

It may record an intended downstream use where that use influences the required simulation design.

However:

Intended Use ≠ Authorized Reliance

Formal validation and reliance questions belong to VALIDOS® and other applicable governance layers.

Simulation Purpose vs Validation Purpose

SIMULOS® and VALIDOS® must remain separate.

SPSS asks:

Why are we performing this simulation and what must it cover?

VALIDOS® VPSS asks:

Why are we validating this Validation Object and what must that validation establish?

A simulation may later become: validation evidence, a Validation Object, part of a Validation Method, or one input into a broader validation determination.

The purposes may interact, but they are not the same governance question.

Simulation Depth

Simulation Depth defines how deeply the governed simulation inquiry must examine the selected Simulation Object characteristics and conditions.

Depth may increase where the simulation purpose involves: high consequence, high complexity, high uncertainty, autonomous behavior, significant

interaction effects, rare failure conditions, safety-critical operations, irreversible decisions, or broad downstream dependence.

Simulation Depth should remain proportional to the purpose. Simulation Depth $\neq$ Validation Depth

SIMULOS® establishes:

Simulation Depth $\neq$ Validation Depth

Simulation Depth governs the required depth of the simulation inquiry.

Validation Depth under VALIDOS® governs the strength and assurance level required to establish a validation conclusion.

One may influence the other without becoming the same concept.

Simulation Precision

Simulation Precision defines the level of detail, granularity, resolution, or quantitative specificity required to satisfy the Simulation Purpose.

Precision may concern: temporal resolution, spatial resolution, behavioral granularity, component detail, parameter resolution, scenario resolution, measurement granularity, or output specificity.

Greater precision is not automatically better.

No False Precision Rule

SIMULOS® establishes:

Simulation precision must not exceed what the simulation purpose, object definition, representation basis, assumptions, inputs, and model structure can meaningfully support.

A highly precise numerical output does not automatically imply highly precise representation of reality.

Purpose-Depth Alignment

Simulation Depth should be sufficient for the Simulation Purpose.

A mismatch may occur where:

High-Consequence Purpose → Low Simulation Depth

or:

Exploratory Purpose → Excessive Simulation Burden

SPSS governs alignment rather than maximum complexity.

Purpose-Precision Alignment

Simulation Precision should also be appropriate to purpose.

A simulation intended only to identify broad directional behavior may not require the same precision as one intended to examine a narrow engineering tolerance.

Scope-Depth Alignment

Depth must apply to the actual Simulation Scope.

A simulation cannot compensate for missing scope merely by examining the included portion in extreme detail.

Therefore:

Deep Simulation of Narrow Scope $\neq$ Broad Simulation Coverage Scope-Precision Alignment

High precision within one narrow scope does not justify claims concerning excluded conditions.

This prevents precise outputs from creating artificial scope credibility.

Consequence Context

The consequence attached to the intended simulation use may influence required: depth, precision, scenario breadth, representation fidelity, documentation, or later validation.

SPSS may therefore record a Simulation Consequence Context where materially relevant.

It does not replace general risk governance.

Exploratory Simulation

An exploratory simulation may intentionally operate with: broader uncertainty, lower precision, incomplete knowledge, wider hypothesis space, or provisional assumptions.

This is legitimate if the Simulation Purpose clearly identifies the exploratory nature of the inquiry.

Exploratory results must not silently become high-assurance conclusions.

Decision-Support Simulation

Where simulation is intended to support a decision, the decision context should be sufficiently described to establish the required scope, depth, and precision.

SPSS does not govern the decision itself.

It governs the simulation inquiry supporting that decision.

High-Consequence Simulation

Where simulation may materially influence: safety, critical infrastructure, autonomous operation, medical decisions, aerospace operations, mission-critical systems, financial stability, or other high-consequence environments,

SPSS should require correspondingly explicit purpose, scope, exclusions, depth, and applicability.

Simulation Purpose Change

The Simulation Purpose may change during the lifecycle.

A Material Purpose Change occurs where the new purpose would require materially different: representation, assumptions, model structure, inputs, scenarios, execution, outputs, or correlation requirements.

Material purpose change requires downstream reassessment.

Simulation Scope Change

Simulation Scope may also change.

Relevant changes include:

Scope Expansion

Scope Contraction

Scope Recomposition

Scope Extension

Scope Restriction

or:

Applicability Change

Material scope change must remain traceable.

Purpose & Scope Change Propagation

SIMULOS® establishes:

A material change to Simulation Purpose or Scope must propagate to every downstream simulation governance space materially dependent upon that purpose or scope.

Potentially affected spaces include: Reality Representation, Simulation

Assumptions, Model & Environment, Inputs & Data, Scenario Design, Simulation Execution, Simulation Output Governance, Simulation-to-Reality Correlation, Simulation Evidence & Reliance Boundary, and Simulation Lifecycle & Resimulation.

No Retrospective Purpose Redefinition Rule

A Simulation Purpose must not be materially redefined after results become known merely to make those results appear more successful or relevant.

Material change must be explicitly documented and affected simulation governance reconsidered.

No Retrospective Scope Redefinition Rule

Likewise, the scope must not be silently changed after execution so that failed or unmodeled conditions disappear from the represented inquiry.

Purpose & Scope Traceability

The simulation lifecycle should remain traceable to:

Simulation Object → Simulation Purpose → Simulation Questions → Simulation Scope → Inclusions → Exclusions → Applicability → Depth → Precision

This relationship allows later reviewers to determine what the simulation was actually designed to establish.

Simulation Purpose & Scope Record

SPSS should produce a persistent Simulation Purpose & Scope Record sufficient to reconstruct: why the simulation was performed, which questions it addressed, what was included, what was excluded, where the simulation applied, how deeply it was intended to examine the object, and what level of precision was required.

Operational Space

SPSS governs: Simulation Purpose, primary and secondary purposes, Simulation Questions, Simulation Scope, scope inclusions, scope exclusions, applicability context, partial scope, scope fragmentation, Simulation Depth, Simulation Precision, consequence context, intended simulation use, purpose changes, scope changes, scope limitations, purpose and scope traceability, and the persistent Simulation Purpose & Scope Record.

Governance Boundary

SPSS begins when:

a governed Simulation Object exists and requires a defined reason, inquiry boundary, and simulation extent.

SPSS ends when:

Simulation Purpose, Simulation Questions, Scope, material inclusions and exclusions, Applicability Context, Simulation Depth, and required Simulation Precision are sufficiently established to support governed Reality Representation.

Governance Scope

SPSS governs only the purpose and scope of simulation.

It does not govern: Simulation Object identity, Simulation Object boundary itself, reality-representation fidelity, simulation assumptions, model construction, simulation environment construction, input-data suitability, scenario design, execution, outputs, simulation-to-reality correlation, formal validation, Validation State, operational reliance authorization, general integrity, risk governance, accountability, or audit execution.

These responsibilities belong to other SIMULOS® Parent Standards or adjacent OOF® architectures.

Minimum Governance Requirements

A conforming application of SPSS should establish:

1. an explicit Simulation Purpose, 2. a clear relationship between purpose and the governed Simulation Object, 3. one or more identifiable Simulation Questions, 4. a bounded Simulation Scope, 5. explicit material scope inclusions, 6. explicit material scope exclusions, 7. a defined Simulation Applicability Context, 8. sufficient Simulation Depth relative to purpose, 9. appropriate Simulation Precision, 10. visibility of intended downstream use where materially relevant, 11. prevention of purpose substitution, 12. prevention of scope inflation, 13. prevention of retrospective purpose or scope redefinition, 14. traceability of material purpose and scope changes, 15. and a persistent Simulation Purpose & Scope Record.

Governance Outputs

SPSS may produce: Simulation Purpose Statement, Primary Simulation Purpose, Secondary Simulation Purpose Record, Simulation Question Set, Simulation Scope Definition, Simulation Scope Map, Scope Inclusion Record, Scope Exclusion Record, Partial Scope Record, Scope Fragmentation Map, Simulation Applicability Context, Simulation Depth Requirement, Simulation Precision Requirement, Simulation Consequence Context, Intended Simulation Use Record, Purpose Limitation Record, Scope Limitation Record, Purpose Change Record, Scope Expansion Record, Scope Contraction Record, Scope Recomposition Record, Purpose & Scope Change Assessment, Purpose & Scope Traceability Map, and Simulation Purpose & Scope Record.

Operational Applications

Simulation Purpose & Scope Standard may be applied across: artificial intelligence, autonomous systems, robotics, aerospace, space missions,

aviation, autonomous vehicles, manufacturing, digital twins, scientific simulation, engineering simulation, healthcare, financial systems, critical infrastructure, environmental systems, mission planning, system-of-systems simulation, policy simulation, organizational simulation, population simulation, training environments, safety analysis, failure analysis, and future operational ecosystems.

Relationship to Other OOF® Architectures

Simulation Purpose & Scope Standard interoperates with: GOA™ ORA™ OBIDENITY® VALIDOS® INTEGROS® ADIT® AGA™ AIG® CLIA® MGIA™ ASGA™ RIS™

by establishing the simulation-specific inquiry boundary that downstream simulation governance consumes.

GOA™ may govern broader governance authority and institutional scope.

ORA™ may provide the operational-reality context that the simulation intends to examine or represent.

VALIDOS® may establish a separate Validation Purpose & Scope where simulation outputs, models, or simulation-derived claims enter formal validation.

RIS™ may influence required Simulation Depth, consequence context, and later scenario intensity.

AGA™ may establish responsibility for simulation-purpose decisions and scope authorization.

ADIT® may preserve evidence showing which Simulation Purpose & Scope governed a simulation at a particular point in time.

INTEGROS® may govern integrity of the underlying scope and purpose records.

Specialized AI, cognition, memory, and autonomous-system architectures may contribute domain conditions that influence what the simulation must examine without replacing SPSS.

SPSS governs specifically:

why the simulation exists and what the simulation inquiry is permitted to claim it covers.

Architectural Position

Simulation Purpose & Scope Standard (SPSS) is the second Parent Standard of SIMULOS® — Simulation Governance Architecture.

The architecture begins:

Simulation Object

↓

Simulation Purpose & Scope

SOS establishes:

What exactly is being simulated?

SPSS establishes:

Why is it being simulated, what must the simulation cover, what is excluded, and how deeply must the inquiry operate?

Once those conditions are sufficiently established, the next architectural question becomes:

Which aspects of reality must now be represented inside the simulation, and at what level of abstraction and fidelity?

This transition leads directly into:

Reality Representation Governance

Foundational Principle

A simulation cannot legitimately answer a broader question than the purpose and scope for which it was actually designed.

Canonical Closing Statement

Simulation Purpose & Scope Standard (SPSS) establishes the inquiry boundary of SIMULOS® by governing why a simulation is performed, which questions it is intended to answer, what portions of the Simulation Object and surrounding conditions it covers, what remains explicitly excluded, where its applicability begins and ends, and what depth and precision are required for the intended simulation purpose. Through purpose definition, scope governance, inclusion and exclusion control, applicability bounding, depth and precision alignment, anti-inflation rules, change propagation, and persistent traceability, SPSS prevents technically successful simulations from acquiring purposes, meanings, or coverage that they were never designed to support. By transforming a governed Simulation Object into a bounded simulation inquiry, SPSS establishes the second methodological foundation of the SIMULOS® lifecycle and prepares that inquiry for governed representation of reality.

Module Architecture

→ Simulation Purpose Definition Module (SPDM)

→ Simulation Scope Module (SSCM)

→ Simulation Exclusion Module (SEXM)

→ Simulation Applicability Context Module (SACM)

→ Simulation Depth & Precision Module (SDPM)

Related Documents

→ [About Simulation Purpose & Scope Standard \(SPSS\)](#)

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>